

# Chapter 1.3 Random Variables

Rick Durrett Probability: Theory and Examples

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Built from the local Chapter 1 LLMwiki flash cards.

# Roadmap for 1.3

## Learning goal

Prove measurability quickly using generators and closure under limits/composition.

1. Random variables as measurable maps.
2. Generator checking and inverse images.
3. Arithmetic, composition, and limits of random variables.
4. Semicontinuity and discontinuity sets.
5.  $\sigma(X)$  factorization: when  $Y = f(X)$ .

# Random Variables as Measurable Maps

## Definition

A random variable is a measurable map

$$X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

Equivalently,

$$\{X \in B\} \in \mathcal{F} \quad \text{for every } B \in \mathcal{B}(\mathbb{R}).$$

## Pitfall

A random variable is not the same thing as its distribution. Different maps may have the same law.

**LLMwiki:** Random variables as measurable maps; ID: durrett\_ch1\_measurable\_map

# Checking Measurability on Generators

## Theorem

If  $\mathcal{S} = \sigma(\mathcal{A})$ , then to prove  $X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{S})$  is measurable, it is enough to check

$$X^{-1}(A) \in \mathcal{F} \quad \text{for every } A \in \mathcal{A}.$$

Let

$$\mathcal{D} = \{B \in \mathcal{S} : X^{-1}(B) \in \mathcal{F}\}.$$

Since inverse images preserve complements and countable unions,  $\mathcal{D}$  is a sigma-field. If  $\mathcal{A} \subseteq \mathcal{D}$ , then  $\sigma(\mathcal{A}) \subseteq \mathcal{D}$ , so all target measurable sets have measurable inverse image.

**LLMwiki:** [Checking measurability on generators](#); ID: durrett\_ch1\_measurability\_generator

# Composition of Measurable Maps

## Theorem

If  $X$  is measurable and  $f$  is measurable, then  $f(X)$  is measurable.

For any Borel set  $B$ ,

$$\{f(X) \in B\} = \{X \in f^{-1}(B)\}.$$

Since  $f$  is measurable,  $f^{-1}(B)$  is Borel; since  $X$  is measurable, the event on the right is in  $\mathcal{F}$ .

**LLMwiki:** [Composition of measurable maps](#); ID: `durrett_ch1_composition_measurable`

# Arithmetic of Random Variables

## Theorem

Finite sums, products, maxima, minima, and continuous functions of random variables are random variables.

- Vector trick: first prove  $(X, Y)$  is measurable.
- Then compose with continuous maps such as  $(x, y) \mapsto x + y$  or  $(x, y) \mapsto xy$ .
- Direct trick for sums: use rational cuts.

**LLMwiki:** Arithmetic of random variables; ID: durrett\_ch1\_measurable\_arithmetic

# Limits Preserve Measurability

## Theorem

If  $(X_n)$  are measurable real functions, then

$$\sup_n X_n, \quad \inf_n X_n, \quad \limsup_n X_n, \quad \liminf_n X_n$$

are measurable. Any pointwise limit of measurable functions is measurable.

Use countable descriptions such as

$$\left\{ \sup_n X_n \leq a \right\} = \bigcap_{n=1}^{\infty} \{X_n \leq a\}.$$

Then write  $\limsup/\liminf$  through countable suprema and infima:

$$\limsup_n X_n = \inf_m \sup_{n \geq m} X_n.$$

# $\sigma(X)$ Factorization

## Fact

A random variable  $Y$  is  $\sigma(X)$ -measurable if and only if there is a Borel measurable  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$Y = f(X).$$

- Easy direction: inverse images compose.
- Hard direction: approximate  $Y$  by simple functions or dyadic bins.

**LLMwiki:** Functions measurable with respect to  $\sigma(X)$ ; ID: durrett\_ch1\_sigma\_x\_factorization



## Exercise 1.3.1

Show that if  $\mathcal{A}$  generates  $\mathcal{S}$ , then

$$X^{-1}(\mathcal{A}) \equiv \{\{X \in A\} : A \in \mathcal{A}\}$$

generates

$$\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}.$$

**LLMwiki:** [Checking measurability on generators](#); ID: durrett\_ch1\_measurability\_generator

## Exercise 1.3.1: Proof Skeleton

Let  $\mathcal{G} = \sigma(X^{-1}(\mathcal{A}))$ . Since  $\mathcal{A} \subseteq \mathcal{S}$ ,  $\mathcal{G} \subseteq \sigma(X)$ . For the reverse inclusion, define

$$\mathcal{D} = \{B \in \mathcal{S} : \{X \in B\} \in \mathcal{G}\}.$$

The class  $\mathcal{D}$  is a sigma-field and contains  $\mathcal{A}$ , hence  $\mathcal{D} = \mathcal{S}$ . Therefore every  $\{X \in B\}$  with  $B \in \mathcal{S}$  lies in  $\mathcal{G}$ .

**LLMwiki:** [Checking measurability on generators](#); ID: durrett\_ch1\_measurability\_generator

## Exercise 1.3.2

Prove Theorem 1.3.6 when  $n = 2$  by checking directly that

$$\{X_1 + X_2 < x\} \in \mathcal{F}.$$

**LLMwiki:** Rational cut proof for sum measurability; ID: `exercise_method_rational_cut_sum_measurability`

## Exercise 1.3.2: Proof Skeleton

For fixed  $x \in \mathbb{R}$ , write

$$\{X_1 + X_2 < x\} = \bigcup_{q \in \mathbb{Q}} (\{X_1 < q\} \cap \{X_2 < x - q\}).$$

The equality follows from density of  $\mathbb{Q}$  between  $X_1(\omega)$  and  $x - X_2(\omega)$ . The right side is a countable union of measurable events.

**LLMwiki:** Rational cut proof for sum measurability; ID: `exercise_method_rational_cut_sum_measurability`

## Exercise 1.3.3

Show that if  $f$  is continuous and  $X_n \rightarrow X$  almost surely, then

$$f(X_n) \rightarrow f(X)$$

almost surely.

**LLMwiki:** Composition of measurable maps; ID: durrett\_ch1\_composition\_measurable

## Exercise 1.3.3: Proof Skeleton

Let

$$\Omega_0 = \{\omega : X_n(\omega) \rightarrow X(\omega)\}.$$

Then  $\mathbb{P}(\Omega_0) = 1$ . For  $\omega \in \Omega_0$ , continuity of  $f$  gives

$$f(X_n(\omega)) \rightarrow f(X(\omega)).$$

Hence convergence holds on a probability-one set.

**LLMwiki:** [Composition of measurable maps](#); ID: durrett\_ch1\_composition\_measurable

## Exercise 1.3.4

- (i) Show that a continuous function from  $\mathbb{R}^d$  to  $\mathbb{R}$  is measurable from  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$  to  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ .
- (ii) Show that  $\mathcal{B}(\mathbb{R}^d)$  is the smallest sigma-field that makes all continuous functions measurable.

**LLMwiki:** Composition of measurable maps; ID: durrett\_ch1\_composition\_measurable

## Exercise 1.3.4: Proof Skeleton

For (i), inverse images of open sets under continuous functions are open, hence Borel. For (ii), suppose  $\mathcal{G}$  makes every continuous real function measurable. For any closed  $C \subseteq \mathbb{R}^d$ , the distance function

$$d_C(x) = \inf_{y \in C} |x - y|$$

is continuous and

$$C = \{x : d_C(x) = 0\}.$$

Thus all closed sets, hence all Borel sets, lie in  $\mathcal{G}$ .

**LLMwiki:** [Checking measurability on generators; ID: durrett\\_ch1\\_measurability\\_generator](#)



## Exercise 1.3.5

A function  $f$  is lower semicontinuous if

$$\liminf_{y \rightarrow x} f(y) \geq f(x),$$

and upper semicontinuous if  $-f$  is lower semicontinuous. Show that  $f$  is lower semicontinuous if and only if  $\{x : f(x) \leq a\}$  is closed for each  $a \in \mathbb{R}$ , and conclude that semicontinuous functions are measurable.

**LLMwiki:** Lower semicontinuity via closed sublevel sets; ID:  
`exercise_method_closed_sublevel_semicontinuity`

## Exercise 1.3.5: Proof Skeleton

If  $f$  is l.s.c. and  $x_n \rightarrow x$  with  $f(x_n) \leq a$ , then

$$f(x) \leq \liminf_n f(x_n) \leq a,$$

so  $\{f \leq a\}$  is closed. Conversely, if all  $\{f \leq a\}$  are closed and  $\liminf f(y_n) < f(x)$  for some  $y_n \rightarrow x$ , choose  $a$  between them and get a contradiction by closedness. Since  $\{f \leq a\}$  is Borel for every  $a$ ,  $f$  is measurable.

**LLMwiki:** Lower semicontinuity via closed sublevel sets; ID:

`exercise_method_closed_sublevel_semicontinuity`

## Exercise 1.3.6

Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be arbitrary and define

$$f^\delta(x) = \sup\{f(y) : |y - x| < \delta\}, \quad f_\delta(x) = \inf\{f(y) : |y - x| < \delta\}.$$

Show that  $f^\delta$  is lower semicontinuous and  $f_\delta$  is upper semicontinuous. Let

$$f^0 = \lim_{\delta \downarrow 0} f^\delta, \quad f_0 = \lim_{\delta \downarrow 0} f_\delta.$$

Conclude that the set of discontinuities of  $f$  is  $\{f^0 \neq f_0\}$  and is measurable.

**LLMwiki:** Discontinuity set from upper and lower local envelopes; ID:

`exercise_method_oscillation_discontinuity_set`

## Exercise 1.3.6: Proof Skeleton

Show  $\{x : f^\delta(x) > a\}$  is open: if  $f^\delta(x) > a$ , some nearby  $y$  has  $f(y) > a$ , and the same  $y$  remains nearby for all  $x'$  close to  $x$ . Thus  $f^\delta$  is l.s.c.; similarly  $f_\delta$  is u.s.c. The limits  $f^0, f_0$  are measurable. Continuity at  $x$  is equivalent to the small-ball upper and lower envelopes squeezing together:

$$f^0(x) = f_0(x).$$

**LLMwiki:** Discontinuity set from upper and lower local envelopes; ID:

`exercise_method_oscillation_discontinuity_set`

## Exercise 1.3.7

A function  $\phi : \Omega \rightarrow \mathbb{R}$  is simple if

$$\phi(\omega) = \sum_{m=1}^n c_m \mathbf{1}_{A_m}(\omega), \quad A_m \in \mathcal{F}.$$

Show that the class of  $\mathcal{F}$ -measurable functions is the smallest class containing the simple functions and closed under pointwise limits.

**LLMwiki:** Limits preserve measurability; ID: durrett\_ch1\_limits\_measurable

## Exercise 1.3.7: Proof Skeleton

Simple functions are measurable, and pointwise limits of measurable functions are measurable. For minimality, approximate any nonnegative measurable  $f$  by simple functions

$$f_n = 2^{-n} \lfloor 2^n (f \wedge n) \rfloor.$$

Then  $f_n \rightarrow f$  pointwise. For general real  $f$ , apply this to  $f^+$  and  $f^-$  and take differences.

**LLMwiki:** Limits preserve measurability; ID: durrett\_ch1\_limits\_measurable

## Exercise 1.3.8

Use the previous exercise to conclude that  $Y$  is measurable with respect to  $\sigma(X)$  if and only if

$$Y = f(X)$$

for some measurable function  $f : \mathbb{R} \rightarrow \mathbb{R}$ .

**LLMwiki:** Functions measurable with respect to  $\sigma(X)$ ; ID: durrett\_ch1\_sigma\_x\_factorization

## Exercise 1.3.8: Proof Skeleton

If  $Y = f(X)$ , then  $\{Y \in B\} = \{X \in f^{-1}(B)\} \in \sigma(X)$ . Conversely, let  $\mathcal{H}$  be the class of  $\sigma(X)$ -measurable functions representable as  $f(X)$ . It contains simple  $\sigma(X)$ -measurable functions and is closed under pointwise limits. By Exercise 1.3.7, every  $\sigma(X)$ -measurable  $Y$  belongs to  $\mathcal{H}$ .

**LLMwiki:** Functions measurable with respect to  $\sigma(X)$ ; ID: durrett\_ch1\_sigma\_x\_factorization



## Exercise 1.3.9

Give a constructive proof of the previous result. Note that

$$\{\omega : m2^{-n} \leq Y(\omega) < (m+1)2^{-n}\} = \{X \in B_{m,n}\}$$

for some Borel set  $B_{m,n}$ , and define  $f_n(x) = m2^{-n}$  for  $x \in B_{m,n}$ . Show that  $f_n(x) \rightarrow f(x)$  and  $Y = f(X)$ .

**LLMwiki:** Constructive representation of  $\text{sigma}(X)$ -measurable functions; ID:

`exercise_method_constructive_sigma_x_representation`

## Exercise 1.3.9: Proof Skeleton

Partition  $\Omega$  into dyadic bins

$$A_{m,n} = \{m2^{-n} \leq Y < (m+1)2^{-n}\}.$$

Since  $A_{m,n} \in \sigma(X)$ , write  $A_{m,n} = \{X \in B_{m,n}\}$ . Make the  $B_{m,n}$  disjoint if needed, then define

$$f_n(x) = \sum_{m \in \mathbb{Z}} m2^{-n} \mathbf{1}_{B_{m,n}}(x).$$

Then  $0 \leq Y - f_n(X) < 2^{-n}$ , so  $f_n(X) \rightarrow Y$ . Let  $f = \lim f_n$  on the Borel set where the limit exists, and 0 elsewhere.

**LLMwiki:** Constructive representation of  $\sigma(X)$ -measurable functions; ID:  
`exercise_method_constructive_sigma_x_representation`

# Checklist: How to Attack 1.3 Problems

1. To prove measurability, reduce to a generator.
2. For sums, use rational cuts when a direct proof is requested.
3. For limits, rewrite events using countable unions/intersections.
4. For semicontinuity, use closed sublevel sets or open superlevel sets.
5. For  $\sigma(X)$ -measurability, think “information carried by  $X$ ” and prove  $Y = f(X)$ .

**LLMwiki:** [Chapter 1.3 flash cards and exercise method cards](#)

## Mini Practice

1. Prove that  $\max(X, Y)$  is measurable using rational cuts.
2. If  $X_n$  are random variables, show  $\{\lim_n X_n \text{ exists}\} \in \mathcal{F}$ .
3. Show every continuous  $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$  is Borel measurable.
4. If  $Y$  is  $\sigma(X)$ -measurable and  $g$  is Borel, explain why  $g(Y)$  is  $\sigma(X)$ -measurable.